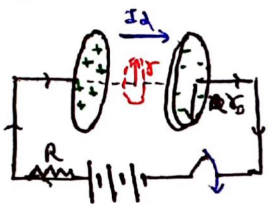
Electromagnetic Wave

Displacement Current :

→ Concept of I_d was proposed in order to maintain 'continuity of current'

→ I_d is the current generated due to change in $\vec{E}$ while charging of capacitor.



Analysis :→

① At any time 't', charge on capacitor is

$$\boxed{q = CV[1 - e^{-t/RC}]}$$

② Conduction current at any time 't'

$$\frac{dq}{dt} = I_c = CV\left[0 + \frac{e^{-t/RC}}{RC}\right]$$

$$\Rightarrow \boxed{I_c = \frac{V}{R}e^{-t/RC}}$$

③ Electric field b/w plates is time varying

$$E_E = \frac{\sigma}{\epsilon_0}$$

$$\Rightarrow \boxed{E_E = \frac{CV[1-e^{-t/RC}]}{\pi r_0^2 \epsilon_0}}$$

④ Electric flux passing through a small area of radius r,

$$\phi_E = E_E \cdot \pi r^2 \cdot \cos 0^\circ$$

$$= \frac{CV[1-e^{-t/RC}]}{\epsilon_0 \cancel{\pi} r_0^2} \times \cancel{\pi} r^2$$

$$\Rightarrow \boxed{\phi_E = \frac{CVr^2}{\epsilon_0 r_0^2}[1 - e^{-t/RC}]}$$

⑤ Although there is no charge b/w capa. but theoritically, we can find it as follows,

$$\therefore \phi_{Total} = \frac{q_{enc}}{\epsilon_0}$$

$$q_{enc.} = \epsilon_0 \cdot \phi_{Total}$$

$$q_{enc.} = \epsilon_0\left(\frac{CVr^2}{\epsilon_0 r_0^2}[1 - e^{-t/RC}]\right)$$

$$\boxed{q_{enc.} = \frac{CVr^2}{r_0^2}[1 - e^{-t/RC}]}$$

Calculation of $I_{displacement}$ ⇒

$$I_d = \frac{dq_{enc.}}{dt}$$

$$\Rightarrow \boxed{I_d = \frac{V}{R}(e^{-t/RC})\left[\frac{r^2}{r_0^2}\right]}$$

So,

$$\boxed{I_d = \epsilon_0\left(\frac{d\phi_E}{dt}\right)}$$

Modification of Ampere's law ⇒

→ When magnetic needle was placed near capacitor, it showed deflection

→ Suggested presence of I_d b/w plates

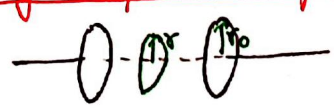
$$\Rightarrow \boxed{\oint B \cdot d\vec{l} = \mu_0[I_c + I_d]}$$

NOTE: In a region, both I_c & I_d can't be present simultaneously. Only one of either wd be present

i.e. when $I_c \neq 0 \Rightarrow I_d = 0$

when $I_d \neq 0 \Rightarrow I_c = 0$

Magnetic field b/w capacitor plates:

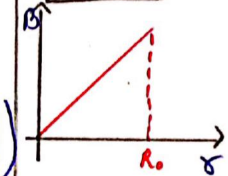


1) When $r < r_0$:→

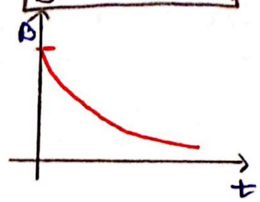
$$B \cdot 2\pi r = \mu_0\left(\frac{Ve^{-t/RC}}{R}\left[\frac{r^2}{r_0^2}\right]\right)$$

$$\Rightarrow \boxed{B = \frac{\mu_0 V}{2\pi R} \cdot e^{-t/RC}\left(\frac{r}{r_0^2}\right)}$$

$\boxed{B \propto r}$



$\boxed{B \propto e^{-t/RC}}$



2) When $r > r_0$:→

→ Here, current can max. pass through area of r_0, S_0

$$B \cdot 2\pi r = \mu_0 \frac{V}{R} e^{-t/RC}\left(\frac{r_0^2}{r_0^2}\right)$$

$$\Rightarrow \boxed{B = \frac{\mu_0 V}{2\pi R} \cdot e^{-t/RC}\left(\frac{1}{r}\right)}$$

$$\Rightarrow \boxed{B \propto \frac{1}{r}}$$

Maxwell's Equation :

1) Gauss law in electro: $\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$

2) Gauss law in mag.: $\oint \vec{B} \cdot d\vec{s} = 0$

3) Faraday's law: emf $= \oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$

4) Maxwell Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0[I_c + I_d]$

Electromagnetic Wave :

→ Doesn't need medium to propagate

→ Produced by accelerated charged particle

Wave No. ($K = \frac{2\pi}{\lambda}$)

Dirn of displacement of particle

$$\boxed{y = A\sin(Kx \pm \omega t)\hat{j}}$$

Disp. of particle

Amplitude

Dirn of prop. of wave

+ ⇒ −ve dirn

− ⇒ +ve dirn

→ Eqn of a travelling transverse wave.

1) $\boxed{v_{wave} = \frac{\omega}{K}}$

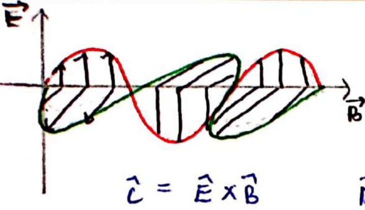
2) $\boxed{\omega = 2\pi\nu = \frac{2\pi}{T}}$

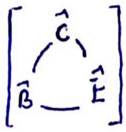
$\boxed{\omega = K \cdot v_{wave}}$

3) $\boxed{K = \frac{2\pi}{\lambda}}$

Properties of EM wave :

1) Dirn of EF, MF and propogation of wave, all are mutually ⊥ar to each other.





$\hat{C} = \vec{E} \times \vec{B}$ $\qquad$ $\vec{B} = \hat{C} \times \vec{E}$

$\vec{E} = \vec{B} \times \hat{C}$

2) EM wave is a transverse wave

3) Both $\vec{E}$ & $\vec{B}$ are in same phase (EK saath max. aur min.)

$$E = E_0 \sin(kx \pm \omega t)$$
$$B = B_0 \sin(kx \pm \omega t)$$

4) EM doesn't req. any medium to propogate. So speed of EM wave

In Vacuum: $\boxed{c = \dfrac{1}{\sqrt{\mu_0 \cdot \varepsilon_0}}}$

In Medium: $\boxed{v = \dfrac{1}{\sqrt{\mu_m \cdot \varepsilon_m}} = \dfrac{1}{\sqrt{\mu_0 \mu_r \cdot \varepsilon_0 \varepsilon_r}}}$

$\boxed{v = \dfrac{c}{\sqrt{\mu_r \cdot \varepsilon_r}}}$

Also,

$\boxed{\text{Refractive Index of med.} = \dfrac{c}{v} = \sqrt{\mu_r \cdot \varepsilon_r}}$

(Generally, Non-magnetic material, $\mu_r = 1$)

5) Ratio of $\vec{E}$ & $\vec{B}$ is constant

V. Imp $\boxed{E_0 = B_0 c}$

6) EM wave carry energy. The rate of flow of energy crossing a unit area is described by Poynting vector $\vec{S}$

$$\boxed{\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})}$$

7) Energy Densities of EM wave :-

$u_E = \frac{1}{2}\varepsilon_0 E^2$ $\qquad$ $u_B = \dfrac{B^2}{2\mu_0}$

Since, $E = E_0 \sin(kx - \omega t)$ $\;|\;$ $E_{rms} = \dfrac{E_0}{\sqrt{2}}$

$B = B_0 \sin(kx - \omega t)$ $\;|\;$ $B_{rms} = \dfrac{B_0}{\sqrt{2}}$

→ So, avg energy densities are

$$\boxed{\langle u_E \rangle = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0 E_{rms}^2}$$

$$\boxed{\langle u_B \rangle = \frac{B_0^2}{4\mu_0} = \frac{B_{rms}^2}{2\mu_0}}$$

→ Energy of EM wave is equally distributed among EF & MF, so

$$u_E = u_B \Rightarrow \boxed{\frac{u_E}{u_B} = 1}$$

→ So total energy density is,

$$u_T = u_E + u_B$$

$$\Rightarrow \boxed{u_T = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}}$$

8) Intensity of EM wave :-

$$\boxed{I = \frac{P_{bulb}}{4\pi r^2}}$$

Also,

$[\langle u \rangle \times c]$ $\boxed{I = \frac{1}{2}\varepsilon_0 E_0^2 c = \frac{B_0^2 c}{2\mu_0}}$

→ The intensity is defined as the average value of Poynting vector taken over one cycle.

$$\boxed{S_{avg.} = \frac{E_0 B_0}{2\mu_0} = \frac{E_0^2}{2\mu_0 c} = \frac{B_0^2 c}{2\mu_0}}$$

Electromagnetic Spectrum :

Cosmic | γ | X | UV | Visible | IR | Micro | Radio

⟶ Energy ↓ses

⟶ λ ↑ses

Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	<10^{-3} nm	Radioactive decay of the nucleus	-do-